

STANSE: Bug-finding Framework for C Programs

Jan Obdržálek, Jiří Slabý and Marek Trtík

Faculty of Informatics, Masaryk University, Brno, Czech Republic
obdrzalek@fi.muni.cz, slaby@fi.muni.cz, trtikm@mail.muni.cz

Abstract. STANSE is a free (available under the GPLv2 license) modular framework for finding bugs in C programs using static analysis. Its two main design goals are applicability on large software projects like Linux kernel and extendability with new bug-finding techniques with a minimal effort. There are currently three bug-finding algorithms implemented within STANSE: `AUTOMATONCHECKER` checks properties described in an automata-based formalism, `THREADCHECKER` detects deadlocks among multiple threads, and `REACHABILITYCHECKER` looks for unreachable code. STANSE has been tested on the Linux kernel, where it has found dozens of previously undiscovered bugs.

1 Introduction

Bug-finding based on static analysis have finally come of age in the last decade. One of the papers to really stir interest was [2], showing that static analysis can efficiently find many interesting bugs in a real-world code, as demonstrated on the Linux kernel. This work eventually led to a successful commercial tool called `COVERITY` [6]. Over the years, several other successful tools, like `CODESONAR` [?] or `KLOCWORK` [8], appeared. However, such fully-featured tools are neither free to obtain, nor is their code available (e.g. for developing new algorithms or tailoring the existing tools to specific tasks). The existing free tools are usually severely limited in what they can do (e.g. `UNO` [10], `SPARSE` [9], `SMATCH`). One notable exception is `FINDBUGS` [4, 7], a successful tool working on Java code.

STANSE is intended to fill this gap for C language. It can be seen in two ways:

1. STANSE is a robust framework (written predominantly in Java) for implementing diverse static analysis algorithms. An implemented algorithm can be immediately evaluated on large real-world software projects written in C as the framework is capable to process such projects (for example, it can process the whole Linux kernel). An implementation of such an algorithm within the framework is called *checker*.
2. As STANSE contains three checkers already, it can be also seen as a working static analysis tool.

The rest of the paper is structured as follows. Section 2 describes the functionality provided by the framework, while Section 3 is devoted to the three

existing checkers. Section 4 presents some results of STANSE on the Linux kernel. The last section summarizes the basic strengths of STANSE and mentions some directions of its future development.

2 Framework Functionality

The functionality provided by the STANSE framework can be separated into several stages. First, the framework reads in the configuration including source file list and checkers to be used. The source files are then parsed and encoded in *abstract syntax tree (AST)*, *control-flow graph (CFG)*, and *call-graph* internal structures. Next, checkers are instantiated, run, and error reports collected and presented.

co se myslí tím “checkers are instantiated”?

Vytvoří se jejich instance. Doslova, vytvoří se pomocí klíčového slova new, aby existoval v paměti.

2.1 Selecting and Parsing Source Files

There are several ways to tell STANSE what source files to analyze. Besides the standard possibilities (a single given file, all files from a given directory, all files listed in a given file), STANSE can also derive all the necessary information from a project Makefile. In the C language family case, it also remembers compiler flags for preprocessing purposes. This functionality was inspired by the SPARSE [9] tool.

STANSE can process source files written in C, more specifically the ISO/ANSI C99 standard and most of the GNU C extensions.¹ Before parsing, the sources are preprocessed by the standard GNU C preprocessor. The code is then parsed into three internal structures: AST, CFG, and call-graph. All these structures can be dumped in a textual or graphical form.

Ty interní struktury odpovídají procedurám nebo modulům? A platí modul=file?

Jak je psáno níže, modul tedy odpovídá compilation unit, což je jeden soubor. CFG odpovídá jedné funkci, AST je rozkouskováno po uzlech CFG a call-graph se (momentálně – kdybysme neorezovali Configuration jen na jeden soubor, tvořil by se call-graph pro všechny v Configuration) tvoří v rámci jednoho souboru. Když se parsuje soubor, tak pro všechny funkce se vyrobí CFG s odpovídajícími AST uzly a také se vyrobí call-graph všech funkcí.

Jak to přesně funguje – smazane to opravdu někdy nepoužívané struktury?

Ano.

Co se myslí tím “transparent”? Nebylo by lepší, kdyby o tom checker nevedel (a tudíž by to bylo “hidden”)?

Jak rozumíš pojmu transparentní? Mám za to, že je to něco, o čem checker nemá ponětí, že existuje. V tomto kontextu mi to zní lepe. Nevím, jak Marek (který to, myslím, napsal), ale já jsem se s hidden v tomto kontextu nesetkal.

As mentioned earlier, the design of the framework is geared towards analyzing large software projects. Such projects consist of hundreds or thousands of modules (compilation units). In practice, it is often impossible to store all the corresponding internal structures in the memory. The STANSE framework therefore applies automatic *streaming* of the internal structures, currently on a module basis. Instead of parsing all source modules at the beginning, a module is streamed in only when STANSE needs to access some internal structures corresponding to the module. In other words, the internal structures are constructed in a lazy manner – we call them *lazy internals*. If the memory occupied by internal structures exceeds a given limit, some internal structures have to be freed before another module is streamed in. The implemented least-recently-used (LRU) algorithm discards the structures of a module if they are not accessed for a significant time. If the discarded structure is later accessed again, the corresponding module is streamed back in. Both lazy internals and streaming are completely transparent to checkers.

¹ In principle, the STANSE framework can handle any programming language for which a frontend is written. We are currently working on C++ support.

2.2 Traversing the Parsed Code

Although all the internal structures are ready to be fully used by a checker at this point, the checker would have to walk through the structures on its own. To prevent unnecessary reimplementations, the most important and heavily used traversal methods are implemented directly inside the framework. Using this functionality, a new checker can be implemented by specifying

- whether it should go through the structures forwards or backwards, breath-first or depth-first,
- if the interprocedural walk-through should be performed or not, and
- methods to be called for each visited node in CFG or AST.

This makes implementation of new algorithms extremely simple.

Tady mam vic dotazu:

- V cem spociva podpora interproceduralni analyzy? V tom, ze si muzu rict, ze do volanych struktur bud vlezu, nebo budu volani ignorovat?
 - JS: rekl bych, ze ano.
- Nemel bych nekde rict, kterou strukturu chci primarne prochazet (call-graf, AST, CFG)?
 - JS: prochazet se da jen CFG. U AST to asi nema smysl, u call-graphu asi ano, ale na to neni pruchodce (nikdo ho zatim nepotreboval).
- Kdyz nahraju modul/soubor, kde ta analyza zacne? Projdu kazdou proceduru?
 - JS: ano, AC prochazi kazdou, TC si vybira ty, ktere nejsou volany nikym jinym v ramci souboru (tzv. pocatecni funkce)
- Predpokladam, ze interni struktury jsou provazany: treba uzly v CFG maji vazbu na odpovidajici podstromy v AST. Je to tak?
 - JS: uplne presne
- Prilepime sem neco jako: *Note that the checkers are independent of each other, they can be executed sequentially or in parallel.* A kdyz mam vic checkeru, to se pro kazdej checker zvlast vytvari internal structures? A noveji checker se musi do toho frameworku nejak registrovat?
 - JS: Note dava smysl.
 - JS: Internals jsou spolecne pro vsechny a jsou pro checkery read-only.
 - JS: registrace checkeru spociva v pridani jednoho radku do CheckerFactory – ano, musi se “registrovat”

Nepatri sem i call-grafy?

Pokud vim (dival jsem se ted do kodu), tak mame pruchodce jen pro CFG.

2.3 Processing Error Traces

Once a checker finds an error, it reports back a set of commented *error traces* (paths in the analyzed code) demonstrating this error. STANSE provides several possibilities to present the error traces: they can be printed to the console, displayed using a built-in error trace GUI browser (see Figure 1), or saved to an external file in XML format. This XML file has a wide variety of possible applications. For example, we supply an SQLITE database creator complemented with a web interface.

The last but not least, STANSE provides an interface for IDE plug-in to be added as an extra module.

Obsahuje stanse nejakou podporu pro slucovani tracu popisujicich stejnou chybu?

JS: ano, slucuje stejne chyby (checker, typ chyby) se stejnym koncovym nebo pocatecnim bodem.

Tady to zni, jako by soucasti STANSE byl i ten web interface. Co presne dela ten creator? Nestaci napsat “For example, we supply a tool translating the XML file into an SQLITE database.”?

JS: Taky, ze ano, primitivni, ale je :). Viz <http://decibel.fi.muni.cz/~xslaby/stanse/>. Creator dela presne ten preklad XML do db, jak pises v te vete.

Te posledni vete nerozumim vubec. Muzete mi to

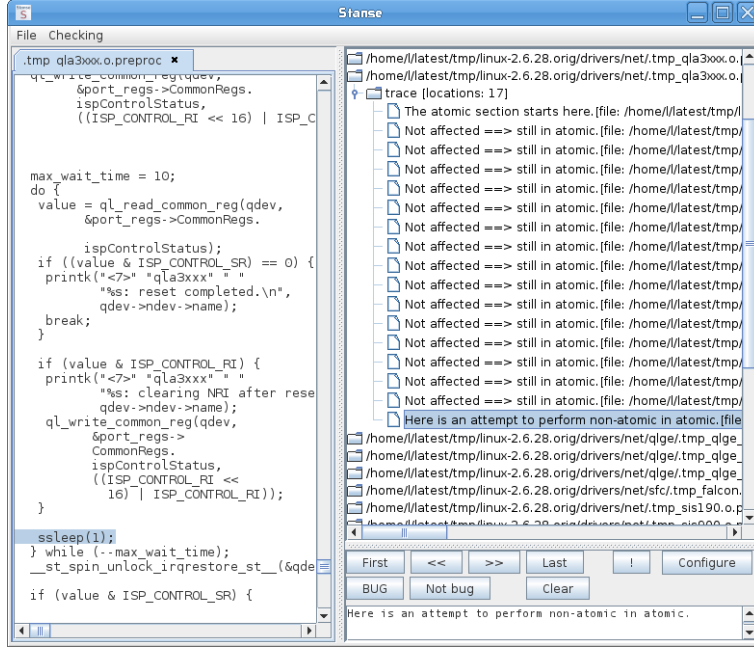


Fig. 1. The error trace GUI browser.

3 Checkers

In this section we describe the three currently available checkers.

AutomatonChecker is heavily influenced by [2]. It takes, as an input, a set of finite-state automata that describe some property which we want to check. Properties like locking discipline, interrupt management, null pointer dereference, dangling pointers and many others can be handled this way.

TADY!!!

Compared to the implementation described in [2] and [3], the technique used differs in several aspects. Here is a list of the most important differences: (1) we do not use metacompilation, (2) automata are not input-language specific, a pattern matching is used instead, (3) our analysis is path-sensitive on checker level, but for this reason (4) the interprocedural analysis is done in the context of a single input file.

The patterns in the configuration are specified in the same language as AST is represented, so currently in XML. The idea behind was to isolate the patterns from the source code.

ThreadChecker aims to check for possible deadlocks in concurrent programs. The technique used is based on the notions of locksets of [5] and deadlock de-

tection by looking for cycles in resource allocation graphs (RAG). THREAD-CHECKER first tries to identify the parts of the code which can run in parallel, as different threads, then it builds a set of *dependency graphs* for each such thread. A dependency graph statically represents the possible locksets during one execution of a thread. Dependency graphs are then combined and all cycles examined, by building the corresponding RAGs.

ReachabilityChecker searches the internals, a CFG more concretely, for unreachable nodes. These are then reported as warnings or errors, depending on importance (e.g. superfluous semicolons are less important than unused branch). This is an easy task and so the writing of such code should be. The code of the checker is below 200 lines including importance decision and much of strings and comments. So it tries to fully demonstrate the power of the STANSE framework and its primary goal is exactly that. The tool is thus very easy to extend, the checker implementors can focus on real techniques.

explicitne zminit, ze to
pouziva traversovani z
frameworku.

4 Results – Linux Kernel

We are exercising the Linux kernel by STANSE since 2.6.28 was released back in 2008. It was the first release we decided to test the usability of STANSE on. One reason to choose the Linux kernel is that it represents a large and freely available codebase, where the absence of bugs is of great concern. Moreover, the Linux kernel fully exercises most features of ANSI C99 and GNU C extensions. The other motivation is to provide a free tool for Linux kernel developers.

Currently, COVERITY PREVENT [6] is daily run on the current development branch of the Linux kernel, and the results are reported back to developers. However, developers kept asking for a tool with which their code will be checked before it appears upstream. Moreover, the kernel code is very specific and the tool should consider these specifics.

We used all three checkers described above. These checkers are not kernel specific and only the configuration for AUTOMATONCHECKER has to be tailor-made to look for the type of bugs commonly found in the kernel code. The properties checked by the AUTOMATONCHECKER in the kernel are:

- correct pairing of functions like balanced locking, check for correct reference counting,
- memory/pointer related bugs like dereferencing null pointers and
- atomic section restrictions to find if someone can cause deadlocks by sleeping inside spinlocks or interrupt handlers.

With this configuration, the running time of STANSE on a common desktop machine with 2 cores and 4 GiB of memory is under 100 minutes. With streaming, the memory usage of the Java process oscillates between 400 and 1000 MiB in case of forced garbage collector executions. The standard STANSE Makefile support was used and we did not have to manually tweak the source code in any way in order to be processed by STANSE.

```

static void untag_chunk(struct node *p) {
    spin_lock(&entry->lock);
    ...
    new = alloc_chunk(size);
    ...
    spin_unlock(&entry->lock);
}
static struct audit_chunk *alloc_chunk(int count) {
    ...
    chunk = kzalloc(size, GFP_KERNEL);
}

```

Fig. 2. Bug found in the audit code

Automaton	Errors	Assorted	Bugs	FP	Ratio
<i>Pairing</i>	266	58	65	143	31.3 %
<i>Memory</i>	86	1	48	37	56.5 %
<i>Atomic</i>	35	1	16	18	47.1 %
Overall	387	60	129	198	39.4 %

Table 1. Errors found by AUTOMATONCHECKER in the 2.6.28 Linux kernel. Assorted = Neroztrizene = nepodarilo se rozhodnout.

In the 2.6.28 release, more than 70 bugs² were reported to kernel developers, most with appropriate patches, and fixed in the next kernel release. Some of these bugs remained undiscovered for seven years³. This is despite the kernel being daily checked⁴ with COVERITY PREVENT and we believe it is due to the filters they use.

Now we present one bug which we found in the code reporting security incidents. An excerpt from the particular code is in the Figure 2. `untag_chunk` functions creates a critical section by a spinlock and calls `alloc_chunk` which in turn can sleep in `kzalloc`. This can, under certain circumstances, lead to deadlock of the whole system. The bug was confirmed and fixed.

The false positive (FP) rate oscillates among the different bug types, the overall ratio is that 40 % of all reported errors are real bugs. This result is quite low, however similarly to [1], it is without any thorough false-positive filtering techniques (cf. Future Work section). The more detailed results about the AUTOMATONCHECKER are in the Table 1, one row per each configuration.

The current kernel versions are checked with other tools like COCCINELLE or SMATCH. Mostly these tools find simple pattern matching style properties like lock-unlock and are successful at that. We continue to check the kernel releases with STANSE too and, thanks to interprocedural analysis, we still are able to find errors on daily basis.

The most convenient advantage on these tools is their availability. The developers or testers can take them whenever they want and check their code instantly,

² A list of confirmed bugs is available at <http://stanse.fi.muni.cz/bugs.html>

³ <http://lkml.org/lkml/2009/3/11/380>

⁴ <http://marc.info/?l=linux-kernel&m=125856033627134&w=2>

because they are freely available and they can handle the kernel code. Further these tools need near no configuration (either because they provide predefined one or there is nothing to configure) and are ready to use out of the box.

5 Conclusions

5.1 Future Work

C++ support, rewrite to C, get rid of XML, daily analysis, results difference. Should this be here? Pay attention, we reference future work section above.

Parser: ... but the parser itself is generated by ANTLR from a top-down C grammar which we developed based on provided example grammar. This is a limiting factor of the whole tool when parsing C sources, because (according to the ANTLR authors) the C parser in Java is unoptimized and slow – this is not true for C parser in C, so we use it for parsing included headers to extract type definitions. The rest of file is handled in Java.

Nowadays, we are working hard on merging a C++ support which is based on CLANG++ from LLVM suite and we are almost finished. We do not use CLANG (at the time of writing this article, 2.8 is stable) for pure C because it still misses some features from the GNU C which are widely used in the Linux kernel (at least local labels and some assembler constraints explicitly).

interni struktury jsou popsány v XML a to je špatné.

There is a plan to integrate STANSE at least into ECLIPSE.

Acknowledgements Jan Kučera is the author of the THREADCHECKER. We would like to thank Linux kernel developers, and Cyrill Gorcunov for STANSE alpha testing and useful suggestions. All authors are supported by the research centre Institute for Theoretical Computer Science (ITI), project No. 1M0545.

References

1. T. Ball, B. Hackett, S. Lahiri, S. Qadeer, and J. Vanegue. Towards scalable modular checking of user-defined properties. *Verified Software: Theories, Tools, Experiments*, pages 1–24, 2010.
2. D. Engler, B. Chelf, A. Chou, and S. Hallem. Checking system rules using system-specific, programmer-written compiler extensions. In *OSDI'00*, pages 1–16, 2000.
3. S. Hallem, B. Chelf, Y. Xie, and D. Engler. A system and language for building system-specific, static analyses. In *PLDI'02*, pages 69–82. ACM, 2002.
4. D. Hovemeyer and W. Pugh. Finding bugs is easy. In *OOPSLA'04*, pages 132–136. ACM, 2004.
5. J.W. Voung, R. Jhala, and S. Lerner. RELAY: static race detection on millions of lines of code. In *ESEC-FSE'07*, pages 205–214. ACM, 2007.
6. COVERITY PREVENT. <http://www.coverity.com/products/>.
7. FINDBUGS. <http://findbugs.sourceforge.net/>.

8. KLOCWORK. <http://www.klocwork.com/products/>.
9. SPARSE. <http://www.kernel.org/pub/software/devel/sparse/>.
10. UNO. <http://spinroot.com/uno/>.